

Review article

Advancing the carbon pricing framework in Indonesia: A systematic review of policies, challenges, and global lessons

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ABSTRACT

Indonesia, the largest GHG emitter and carbon sink reservoir in Southeast Asia, faces the dual challenge of achieving net zero emissions by 2060 while sustaining economic growth. This research examines Indonesia's carbon pricing framework, focusing on two key mechanisms: a modest carbon tax and a nascent ETS. Using an SLR and bibliometric analysis, it evaluates implementation progress, including the introduction of a carbon tax at 2 USD per tonne of CO₂ and an ETS targeting coal-fired power plants, alongside persistent challenges such as limited sectoral coverage, low price signals, and institutional fragmentation. Drawing insights from both emerging economies (e.g., South Africa, Colombia) and advanced systems (e.g., Sweden, Canada), the analysis underscores the importance of phased implementation, strong governance, and international alignment. The paper proposes actionable pathways, including a graduated carbon tax roadmap, staged ETS expansion, and enhanced regional collaboration within ASEAN. This study offers a novel scientific contribution by conducting a multidimensional mapping of Indonesia's carbon pricing literature, which categorizes 65 studies by policy instruments, implementation phases, and sectoral coverage, and by identifying a critical disconnect between academic recommendations and real-world implementation. These findings provide an evidence-based foundation for designing more effective and internationally aligned policy reforms. By aligning these reforms with global best practices and leveraging domestic resources, Indonesia can strengthen its carbon pricing architecture and offer a replicable model for sustainable development in other emerging economies.

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Abbreviations: Greenhouse Gas, (GHG); Enhanced Nationally Determined Contribution, (ENDC); Emission Trading System, (ETS); Association of Southeast Asian Nations, (ASEAN); Research Question, (RQ); Protocol, Search, Appraisal, Synthesis, Analysis, and Report, (PSALSAR); Preferred Reporting Items for Systematic Reviews and Meta-Analyses, (PRISMA); Search, Appraisal, Synthesis, and Analysis, (SALSA); Systematic Literature Review, (SLR); Web of Science, (WoS); Computable General Equilibrium, (CGE); Carbon Capture and Storage, (CCS); Reducing Emissions from Deforestation and forest Degradation plus the sustainable management of forests, and the conservation and enhancement of forest carbon stocks, (REDD+); Carbon Border Adjustment Mechanism, (CBAM); Carbon Economic Value, (CEV); Agriculture, Forestry, and Other Land Use, (AFOLU); Nationally Determined Contribution, (NDC); Gross Domestic Product, (GDP); Long—Term Strategy for Low Carbon and Climate Resilience, (LTS—LCCR); Ministry of Finance, (MoF); Ministry of Environment and Forestry, (MoEF); Ministry of Energy and Mineral Resources, (MEMR); Just Energy Transition Partnership, (JETP); United Nations Framework Convention on Climate Change, (UNFCCC); Low Emission Development, (LED); Partnership for Market Readiness, (PMR); Monitoring, Reporting, and Verification, (MRV); Perform, Achieve, and Trade, (PAT); United Nations Development Programme, (UNDP); European Union, (EU); European Environment Agency, (EEA); North American Free Trade Agreement, (NAFTA); Photovoltaic, (PV); US Dollar, (USD); Singapore Dollar, (SGD); Indonesian Rupiah, (IDR); Canadian Dollar, (CAD); Gigawatt, (GW); Organisation for Economic Co-operation and Development, (OECD); Life Cycle Assessment, (LCA); Convolutional Neural Network, (CNN).

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1. Introduction

The increasing concentration of GHGs in the atmosphere, particularly carbon dioxide (CO₂), is the primary driver of global climate change, posing severe risks to ecosystems, economies, and communities worldwide. Predominantly stemming from industrial activities and fossil fuel combustion, CO₂ emissions exacerbate global warming by trapping heat in the atmosphere, triggering rising sea levels, extreme weather events, and biodiversity loss [1]. The Paris Agreement represents a pivotal global commitment to mitigate these threats, aiming to limit the global temperature rise to 1.5 °C above pre-industrial levels by 2100 [2]. Achieving this target requires coordinated international efforts, with carbon pricing emerging as a crucial instrument for driving emissions reductions and facilitating the transition to a low carbon energy system [3].

Carbon pricing, implemented through mechanisms such as carbon trading and carbon taxes, assigns a monetary value to emissions, creating economic incentives for industries to reduce their carbon footprints [4]. Although this has been widely adopted in developed economies, carbon pricing is gaining momentum in developing nations, where governments face the dual challenge of pursuing economic growth while addressing climate goals [5].

Indonesia provides a compelling and unique case study for understanding carbon pricing in the Global South. As the largest GHG emitter and carbon sink reservoir in Southeast Asia, Indonesia occupies a dual role in the global climate regime, both as a major emitter and as a steward of critical natural resources such as tropical forests and peatlands [6]. It is also Southeast Asia's largest economy and among the top 10 global CO₂ emitters from fossil fuel combustion [7,8]. Moreover, Indonesia is endowed with abundant renewable energy potential, including more than 3294.4 GW in solar, 154.9 GW in wind, 95 GW in hydro, and 23 GW in geothermal, making it one of the most promising developing countries for the clean energy transition [9].

However, Indonesia faces significant complexity in aligning rapid economic growth with ambitious climate mitigation objectives. As a developing nation with a structural reliance on coal and a fragmented institutional landscape, Indonesia must balance developmental priorities with environmental commitments. These characteristics underscore the relevance of studying carbon pricing mechanisms in this context, not only to evaluate national progress but also to offer broader insights for climate policy design in emerging economies.

Indonesia has underscored its commitment to climate action through its ENDC under the Paris Agreement, pledging to reduce emissions by 31.89% unconditionally and 43.2% with international support. The energy sector, Indonesia's largest source of emissions, is a focal point, with targeted reductions of 12.5% unconditionally and 15.5% conditionally [10]. To achieve these ambitious goals, Indonesia has introduced a carbon pricing framework comprising two complementary mechanisms: carbon trading and a carbon tax. These tools are implemented

to reduce emissions while supporting sustainable development and preserving economic competitiveness in a coal-dependent economy with vast renewable energy potential [11].

The ETS is a market-based mechanism designed to control GHG emissions by setting a cap on allowable emissions and enabling the trading of emission permits among regulated entities, ensuring flexibility while maintaining compliance with reduction targets [12,13]. This system promotes cost-effective compliance by allowing entities that emit less than their allocated allowances to sell the surplus to those exceeding their limits. Complementing this, a carbon tax directly imposes a monetary charge on CO₂ emissions, encouraging industries to adopt cleaner technologies [14]. Together, these mechanisms illustrate Indonesia's efforts to balance its climate ambitions with its dependence on coal, which dominates its energy mix, and to support its significant but underutilized renewable energy potential [15].

Despite the growing global adoption of carbon pricing, research has primarily concentrated on developed economies, leaving significant gaps in the understanding of its application in developing countries like Indonesia [16]. Addressing this gap, the present study investigates Indonesia's carbon pricing framework, focusing on its dual approach using an ETS and a carbon tax. Additionally, this work evaluates how these mechanisms align with global best practices and explores lessons that can inform the implementation of carbon pricing in other developing economies.

This research is guided by two key questions:

RQ1: What are the key features of Indonesia's carbon pricing policies, and how do these policies reflect the country's approach to balancing emissions reduction with sustainable development?

RQ2: How does Indonesia's carbon pricing framework compare to those implemented globally, and what lessons can be drawn to inform international carbon pricing strategies?

To achieve these objectives, the PSALSAR framework and PRISMA guidelines are used. The PSALSAR framework ensures a systematic and replicable research process, from defining protocols to synthesizing findings, and PRISMA enhances transparency in literature selection and analysis. These methodologies collectively strengthen the reliability and comprehensiveness of the research.

The structure of this article is outlined as follows. Section 1 introduces the study's rationale, emphasizing the significance of carbon pricing within Indonesia's climate strategy and situating it within the global context. Section 2 describes the research methodology, detailing the application of the PSALSAR framework and PRISMA guidelines for bibliometric analysis and an SLR. Section 3 presents the results of the bibliometric analysis and SLR, focusing on the distinctive features of Indonesia's carbon pricing framework. Section 4 considers the current status of Indonesia's carbon pricing mechanisms, examining their align-

Table 1
Steps, outcomes, and methods in the PSALSAR framework.

Step	Outcomes	Methods
1. Protocol	Define the scope of the study	Review of current carbon pricing landscape in Indonesia
2. Search	Define the search strategy	Scopus and Web of Science databases; Boolean search in abstract, keywords, and title; keywords: (“carbon pricing” OR “carbon tax” OR “emission trading system” OR “cap and trade”) AND (“Indonesia” OR “Indonesian”); data collected until December 2024
3. Appraisal	Study selection and quality assessment	Remove duplicates; apply inclusion criteria; include only English peer-reviewed articles relevant to carbon pricing, carbon tax, ETS, or cap and trade in Indonesia
4. Synthesis	Create database of selected articles	Extract metadata using Biblioshiny
5. Analysis	Code and analyze data	Categorize themes and timelines; summarize trends and findings
6. Report	Communicate results and draw conclusions	Use PRISMA methodology; develop journal article and recommendations

ment with national sustainability goals and providing a comparative analysis of global carbon pricing practices. Finally, Section 5 summarizes the key findings and offers recommendations for policymakers and future research.

2. Systematic literature review

An SLR is a structured, explicit, and reproducible method used for identifying, evaluating, and synthesizing completed and documented work by researchers, scholars, and practitioners [17]. By providing a comprehensive overview of the current state of research, SLRs aim to enhance the understanding of existing scientific knowledge and research contributions within a specific field [18]. This methodology ensures rigor and transparency, making it a valuable approach for addressing complex research questions.

In this research, an SLR was employed to explore the current landscape and perspectives on carbon taxes and ETSs in Indonesia. To guide this review, the PSALSAR framework, a refined adaptation of the SALSA method developed by Grant and Booth [19], was utilized. The PSALSAR framework, introduced by Mengist et al. [20], enhances the SLR process by incorporating an initial protocol phase and a final reporting phase. These additions improve replicability, transparency, and transferability while minimizing potential biases [21,22]. The foundational SALSA method encompasses four steps, namely search, appraisal, synthesis, and analysis, providing a structured metasynthesis approach for systematically evaluating and integrating findings across multiple studies.

Herein, the PSALSAR framework was rigorously implemented in six steps to ensure a thorough and systematic approach. Step 1 defined the study scope by establishing clear objectives and boundaries for the review. Step 2 outlined the search strategy, detailing the selection of databases and search strings to ensure comprehensive coverage. Step 3 involved the appraisal process, assessing the quality and relevance of the retrieved studies. Step 4 focused on synthesizing the data, and Step 5 comprised an in-depth analysis to interpret the findings. Finally, Step 6 compiled and finalized the report. Further details on each stage are provided in Table 1, demonstrating the robustness of the framework in guiding the review process.

2.1. Protocol

Establishing a research protocol is fundamental to conducting an SLR because it ensures the transparency, transferability, and replicability that define a systematic approach [17]. A well-designed protocol provides a structured framework for the review, enabling exhaustive and comprehensive literature searches while minimizing potential biases [20]. In this study, the protocol was applied to investigate carbon pricing mechanisms in Indonesia, focusing on carbon taxes and ETSs.

By systematically examining these key topics, the review aims to offer valuable insights into their mechanisms and implications for Indonesia's climate policy framework.

2.2. Search

The search phase involved identifying and retrieving information sources relevant to this research, which was divided into two main steps: search strategy and delivery [23]. In the search strategy step, appropriate databases were selected, and Boolean search strings were formulated to locate documents within the scope of the SLR. The delivery step then applied these strings to retrieve pertinent literature [18].

Scopus and WoS were selected as the primary databases due to their comprehensive coverage of peer-reviewed journals across multiple disciplines and their stringent indexing standards [24]. These databases are widely acknowledged in bibliometric literature for their curated journal collections, structured metadata, and advanced Boolean operator functionalities, which collectively enhance the quality and reproducibility of systematic reviews [25]. The use of Boolean logic has been shown to increase both the precision and recall of search results in bibliometric analyses, making it a robust tool for targeted literature identification [26].

In accordance with these best practices, this work employed carefully constructed Boolean search strings combining terms such as (“carbon pricing” OR “carbon tax” OR “emissions trading system” OR “cap and trade”) AND (“Indonesia” OR “Indonesian”), applied to document titles, abstracts, and keywords. The Boolean structure was refined iteratively during a pilot phase to ensure the retrieval of seminal literature and minimize irrelevant results. Although formal recall/precision metrics were not calculated, the inclusion of well-known benchmark studies during early-stage testing supports the effectiveness of the search formulation. This search was conducted in December 2024 and yielded a total of 260 relevant documents. The iterative nature of Boolean refinement enabled greater alignment with the study's evolving research focus, thereby increasing the efficiency and accuracy of the literature selection process [27].

Google Scholar was deliberately excluded from the search process. While it offers broad accessibility, it lacks metadata standardization and includes non-peer-reviewed content (e.g., theses, preprints, and grey literature), which can reduce replicability and compromise the bibliometric compatibility required by tools such as Bibliometrix R. To ensure methodological transparency, this exclusion has been explicitly noted and justified in this subsection.

2.3. Appraisal

The appraisal phase was critical for evaluating the search results and identifying relevant papers with valid findings. This process began with

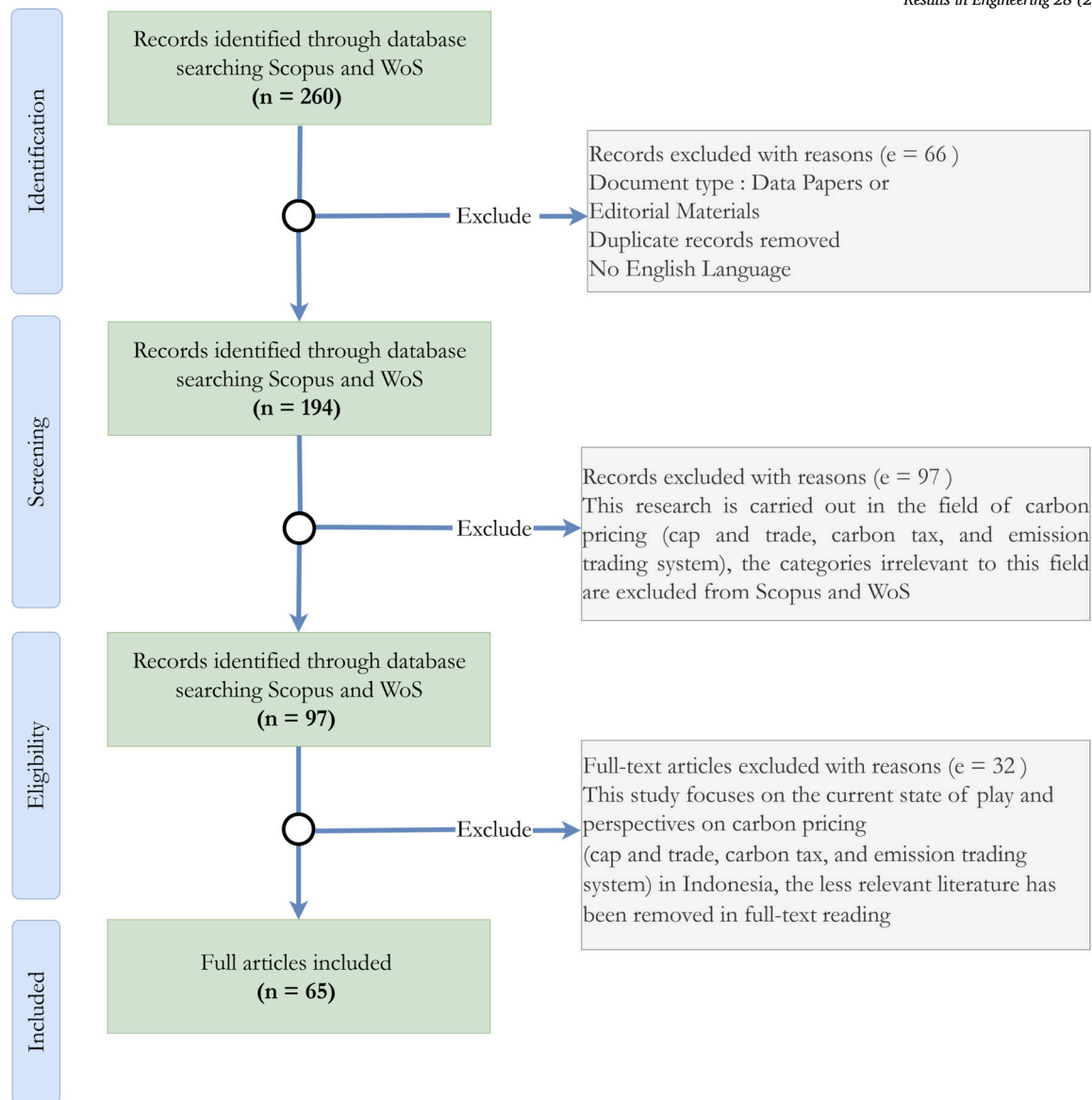


Fig. 1. PRISMA flow diagram of article selection yielding 65 studies on carbon pricing in Indonesia.

an initial screening to ensure their alignment with the review's objectives, following a systematic and repeatable approach guided by predetermined inclusion and exclusion criteria [18]. As outlined in Fig. 1, the first inclusion step required papers to be final publications, written in English, and published in peer-reviewed scientific journals. In this stage, 194 papers were deemed relevant for further analysis. In the subsequent stage, duplicates were removed, abstracts were reviewed, and only papers addressing carbon pricing up to December 2024 were retained. Papers failing to meet these criteria were excluded, resulting in 97 papers for detailed examination. From this pool, 65 papers advanced to the synthesis phase, forming the final database for this study, as depicted in Fig. 1.

To deepen the analysis, Biblioshiny, a bibliometric tool developed in the R programming language, was employed [28]. This application provides a robust suite of tools for importing bibliographic data from databases such as Scopus and WoS and enables comprehensive bibliometric analyses, including the mapping of scientific collaboration, co-word patterns, and co-citation networks [29]. The bibliometric data processing began by refining the database to acquire a targeted dataset spanning the years 1999 to 2024 (see Fig. 2). Keywords extracted from

Scopus and WoS were used to construct a high-quality database that aligns with the objectives of the SLR [30].

Following this, the PRISMA protocol was applied to ensure a structured and transparent evaluation of the research on carbon pricing in Indonesia. As shown in Fig. 1, 65 articles were included in the final analysis, forming the basis for an in-depth assessment of the regulatory and implementation stages of carbon pricing in the country. Eligibility criteria focused on studies closely related to carbon pricing mechanisms, including carbon taxes, and ETSs with a specific emphasis on Indonesia. This rigorous and systematic approach ensured that the final dataset provided a comprehensive and reliable foundation for examining Indonesia's carbon pricing landscape.

2.4. Synthesis

The synthesis phase involved extracting and classifying relevant data from the selected papers to systematically map the evidence base for further analysis [18]. After the selection process, the full texts of the papers were reviewed in detail, and a comprehensive database was created to extract and organize key findings. The data extracted for each study included the year of publication, journal, authors, authors' countries

Table 2
Main information.

Description	Results
Timespan	1999–2024
Journals	62
Book Chapters	2
Review Article	1
Authors	228
Authors of Single-authored Documents	6
Co-Authors per Document	3.77
Average Citations per Document	16.02

and affiliations, research methods, the number of scientific citations, analysis tools, and simulation techniques [31,32]. Additionally, a concise synthesis of the critical outcomes and implications of each paper was prepared, providing a clear summary of its contributions to the field [33].

3. Results

The analysis phase, corresponding to Step 5, involved the evaluation of the content in the selected articles and the extraction of meaningful information, which is presented in this section. This analysis is structured to systematically address the research questions. Specifically, Sections 3.1 and 3.2 focus on answering RQ 1: What are the key features of Indonesia's carbon pricing policies, and how do these policies reflect the country's approach to balancing emissions reduction with sustainable development?

3.1. Bibliometric results

The bibliometric analysis offers a comprehensive overview of the research landscape [34] on carbon pricing in Indonesia, highlighting its scope, trends, key sources, thematic clusters, and central topics. This study provides fresh insights by applying advanced bibliometric tools and thematic analysis to uncover the interdisciplinary and evolving nature of this field. By analyzing the dataset's attributes and visualizing the data (Table 2 and Figs. 2 to 5), the critical strengths, gaps, and opportunities were identified, offering actionable insights for advancing research and policymaking.

Table 2 summarizes the dataset's characteristics, covering the period from 1999 to 2024. Contributions include 62 journal articles, 2 book chapters, and 1 review article, underscoring the academic emphasis on carbon pricing. However, the limited diversity in publication formats indicates an opportunity to expand dissemination through alternative platforms to reach broader audiences. Authorship data reveals significant collaboration, with contributions from 228 authors and only 6 single-authored documents. The average of 3.77 co-authors per document reflects the interdisciplinary nature of carbon pricing research, which spans economics, energy systems, environmental policy, and governance.

The dataset demonstrates substantial academic impact, with an average of 16.02 citations per document. This high citation count signals the relevance of Indonesia's carbon pricing research within the global climate change discourse. The collaborative and interdisciplinary approach has likely contributed to the depth and applicability of findings, making this field a valuable resource for addressing both local and international climate challenges. As Indonesia intensifies its climate commitments, these academic contributions provide a foundation for effective policy design.

Fig. 2 highlights trends in annual and cumulative publications, revealing steady growth in research output over time. From 1999 to 2011, only two publications were recorded, reflecting limited prioritization of carbon pricing during this early period, which coincided with emerging global conversations on market-based mechanisms for emissions reduction. From 2012 to 2017, annual publications gradually increased, with

a cumulative total of 11 by the end of this period, signaling growing interest in carbon pricing as a policy tool.

The most notable growth in research output began in 2018, with annual publications peaking at 18 in 2023. This surge aligns with Indonesia's implementation of carbon pricing policies, particularly its ETS, and the strengthening of its climate commitments under the Paris Agreement. The steady increase in cumulative publications highlights the expanding academic and policy interest in carbon pricing. This upward trend underscores the field's growing maturity and its central role in supporting Indonesia's climate strategy.

Fig. 3 depicts the most influential academic sources, with Energy Policy emerging as the leading journal, contributing five publications focused on policy frameworks and mechanisms. The International Journal of Energy Economics and Policy and the Journal of Cleaner Production have each contributed four publications, emphasizing economic, environmental, and sustainability dimensions. While the dominance of high-impact journals highlights their pivotal role, the analysis also reveals opportunities to diversify publication outlets and engage with a wider audience.

Fig. 4 illustrates the co-occurrence network of research themes related to carbon pricing in Indonesia, providing insights into the interconnectedness and clustering of central concepts. Dominant keywords, such as "Indonesia," "carbon," "emission control," and "climate change," occupy the core of the network, reflecting their central role in discussions on emissions reduction and climate action. The prominence of "Indonesia" underscores the geographic focus, whereas "carbon" and "emission control" emphasize the policy and research priorities aimed at addressing GHG emissions. Additionally, "climate change" serves as a thematic anchor, linking carbon pricing efforts to Indonesia's broader sustainability objectives and global climate commitments. The high degree of connectivity between these keywords reveals the interdisciplinary nature of carbon pricing research, spanning economic, environmental, and energy domains.

The clusters within the network further highlight distinct thematic areas of focus. The green cluster centers on terms like "carbon price," "taxation," and "economic analysis," underscoring the emphasis on financial mechanisms and cost-effectiveness in policy design. In contrast, the blue cluster features terms like "electricity generation," "coal," and "energy," reflecting the critical role of Indonesia's energy sector, particularly its reliance on coal, in achieving emissions reductions. The purple cluster addresses environmental policies, with keywords like "deforestation" and "pollution tax," highlighting the integration of carbon pricing with forest conservation and pollution control efforts. Additionally, the orange and pink clusters focus on terms like "alternative energy" and "investments," demonstrating the growing importance of innovation and financial incentives to support renewable energy and low-carbon technologies. Together, these clusters reveal a multifaceted approach to carbon pricing in Indonesia, integrating economic, energy, and environmental dimensions to balance emissions reductions with sustainable development goals.

Fig. 5 builds on these insights by categorizing the themes using four quadrants, based on their development (density) and relevance (centrality). The Motor Themes quadrant includes "Indonesia," "carbon," and "emission control," which are well-developed and central to the research field. These themes form the backbone of Indonesia's carbon pricing discourse, aligning with its global climate commitments and underscoring the maturity of these topics.

The Basic Themes quadrant highlights foundational but underdeveloped topics such as "emissions trading," "deforestation," and "fossil fuel." These areas are central to the discourse but require further research to address gaps in Indonesia's carbon pricing mechanisms. Expanding research in these areas may enhance the effectiveness of the ETS and carbon tax while addressing emissions from key sectors such as forestry and energy.

The Emerging or Declining Themes quadrant identifies promising but underexplored topics, such as "peatland reduction" and "hydrogen

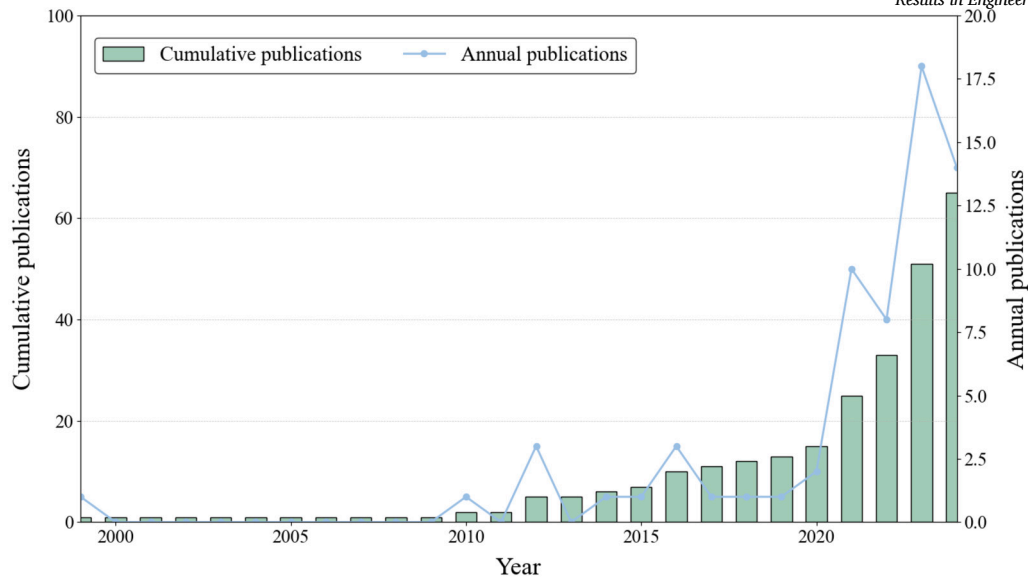


Fig. 2. Annual and cumulative publications on Indonesia's carbon pricing.

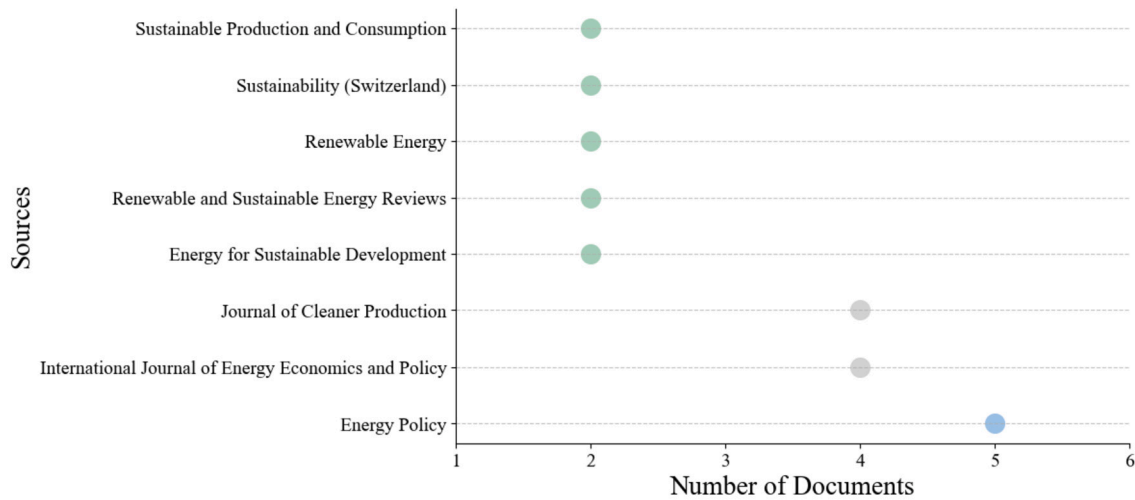


Fig. 3. Most relevant sources.

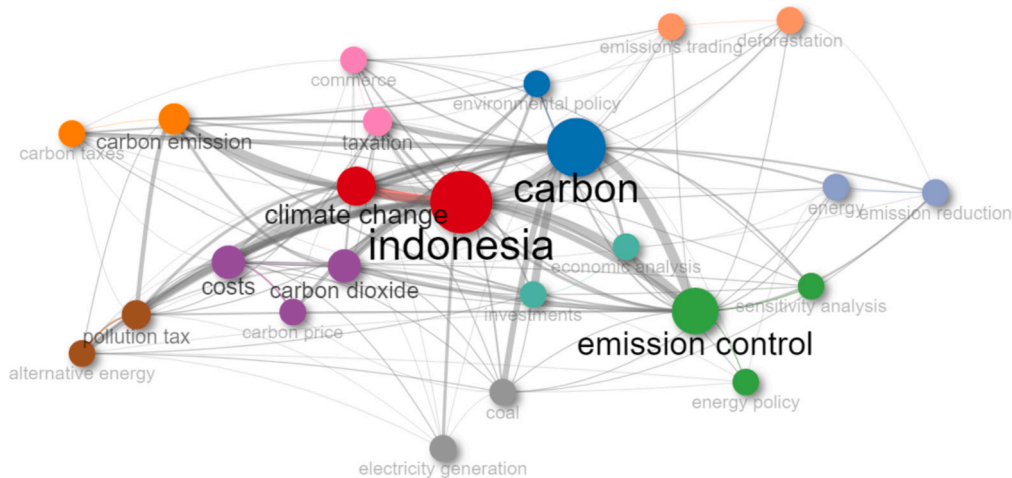


Fig. 4. Co-occurrence network.

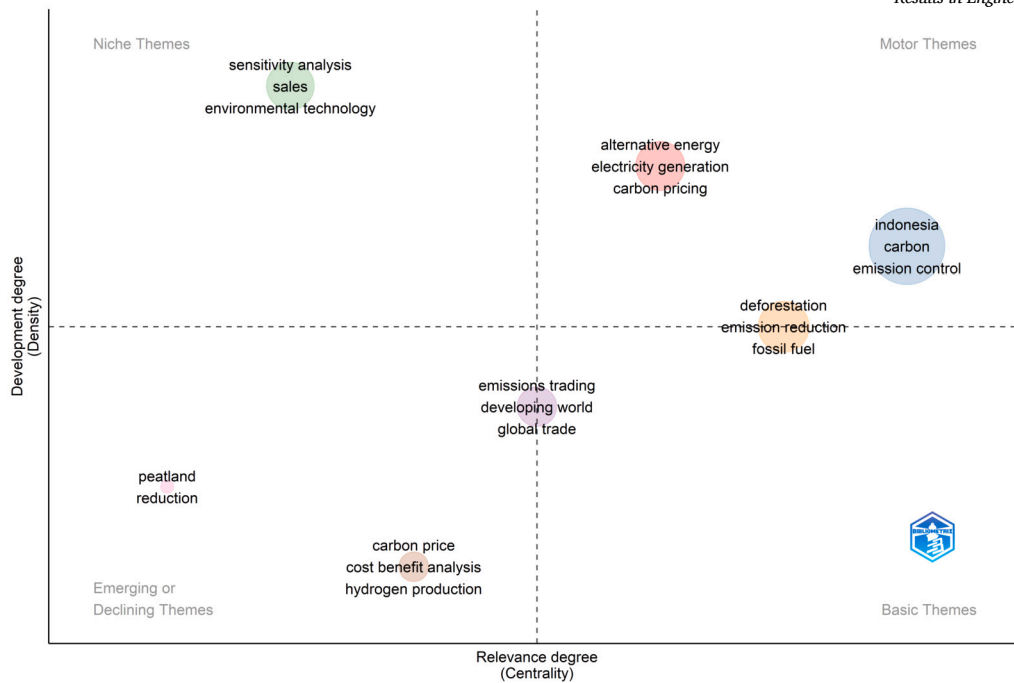


Fig. 5. Thematic map.

production.” These themes offer significant potential, particularly in leveraging Indonesia’s vast peatland reserves as carbon sinks and exploring hydrogen as a renewable energy source. Further research in these areas may complement existing policies and strengthen Indonesia’s transition to a low-carbon economy.

The Niche Themes quadrant features specialized topics like “sensitivity analysis” and “environmental technology.” While these areas are well-developed, they remain peripheral to central discussions. Integrating insights from these themes into broader policy narratives may enhance the technical and practical understanding of carbon pricing mechanisms. Together, Figs. 4 and 5 provide a roadmap for refining Indonesia’s carbon pricing framework by addressing foundational gaps and integrating emerging and niche topics.

The bibliometric results emphasize the dynamic and interdisciplinary nature of the research on carbon pricing in Indonesia, showcasing strong collaboration and growing academic influence. The steady rise in publication output, the prominence of key sources, and the interconnectedness of research themes highlight the increasing importance of this field in addressing Indonesia’s climate goals. By identifying well-established research areas, emerging opportunities, and underexplored gaps, this analysis provides actionable insights for advancing both academic research and practical policymaking.

3.2. Systematic literature review analysis

Tables 3, 4, and 5 summarize the 65 articles analyzed in this study, highlighting key findings on Indonesia’s carbon pricing policies. Carbon pricing plays a critical role in ensuring the smooth operation of the roadmap toward achieving net zero emissions, particularly in the electricity sector. As carbon prices directly impact national economies and environmental sectors, the definitions of carbon trading and carbon tax can be delineated based on the current regulations governing Indonesia’s carbon pricing mechanisms.

1. Carbon trading is a market-oriented approach to reducing GHG emissions at national and international levels by participating in carbon exchange. The procedural framework for carbon trading primarily involves the ETS and greenhouse gas offsets, which apply to different sectors. The carbon trading mechanism involves treating

emission rights as market commodities that are marketed. This approach significantly promotes the reduction of carbon emissions. Regulatory authorities provide emitters (organizations) with free allowances for carbon emissions. If their actual emissions are lower than the allocated allowances, the surplus can be traded for profit on the carbon market. Meanwhile, emitters that release emissions exceeding the permitted allowances must purchase additional allowances [35].

2. Carbon taxes are imposed on products and services that contain or have carbon potential and activities that may cause carbon emissions and prove detrimental to the environment. This mechanism includes central and regional tax impositions, customs and duties, and other state taxes. The collection depends on carbon content, carbon emissions, actual carbon emissions, or climate change mitigation measures. Carbon taxation is an effective strategy to mitigate carbon emissions and encourage consumers to choose products with lower carbon footprints by directly incorporating carbon costs into the pricing structure [36]. As a market-oriented approach that can optimally and cost-effectively reduce carbon emissions and induce market consumption in the short term, carbon tax leads to significant emissions reductions in the long term [37].

To enhance clarity and facilitate synthesis, Table 3, Table 4, and Table 5 group the 65 reviewed articles into three major implementation categories: conceptual/simulation, pilot/early-stage, and legislated. This classification supports policy learning and comparative insights across Indonesia’s carbon pricing evolution. The 65 articles provide a comprehensive view of Indonesia’s carbon pricing framework, emphasizing its dual mechanisms, carbon taxes and ETSs, as integral parts of carbon trading. The findings reveal the importance of integrating these approaches with broader goals for energy transition and sustainability. However, challenges such as weak governance, stakeholder opposition, and limited institutional capacity remain significant. Collectively, this body of work highlights the need for further research to close these gaps and strengthen the overall effectiveness of carbon pricing in Indonesia.

A further analysis of the 65 reviewed articles reveals discernible patterns in the evolution of Indonesia’s carbon pricing discourse. The largest share of studies belongs to the conceptual or simulation-

Table 3

Classification of carbon pricing articles in the conceptual/simulation group.

No.	Author(s)	Policy Instrument	Implementation Phase	Sectoral Coverage	Key Focus
1	Shrestha & Marpaung [38]	Carbon Tax	Conceptual/Simulation	Energy	Carbon taxes enhance efficiency and reduce emissions, with impacts varying by tax level and technology changes.
2	Masseti & Tavoni [46]	ETS	Conceptual/Simulation	Energy	An Asia—specific ETS promotes equitable carbon pricing, lowers abatement costs, and enhances regional cooperation on climate mitigation.
3	Yusuf & Resosudarmo [47]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax in Indonesia has a progressive distributional impact, benefiting rural and low-income households while reducing poverty through revenue recycling.
4	Hasegawa et al. [48]	Carbon Pricing	Conceptual/Simulation	Forestry and Agriculture	Carbon tax incentivizes emissions reductions in AFOLU sectors, supporting Indonesia's NDC targets through cost-effective land-based mitigation strategies.
5	Sujarwo [49]	ETS	Conceptual/Simulation	Forestry	Bamboo forests in Bali serve as effective carbon sinks, supporting emission trading and sustainable community development.
6	Dutu [50]	Implicit Carbon Pricing	Conceptual/Simulation	Energy	Phasing out subsidies and implementing carbon pricing can reduce fossil fuel reliance, promote renewables, and support Indonesia's emissions targets.
7	Urata et al. [51]	Carbon Tax and ETS	Conceptual/Simulation	Industry	Carbon tax drives low-carbon supply chain optimization, aligning cost and emissions reduction through supplier shifts and life cycle inventory-based decisions.
8	Miyata et al. [52]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax reduces CO ₂ emissions but impacts economic growth, with revenue transfers improving household welfare and mitigating negative effects.
9	Wibert et al. [53]	Carbon Tax	Conceptual/Simulation	Energy	Targeting a carbon tax on coal lowers GHG emissions, enhances welfare, and promotes geothermal energy, economically outperforming a broad fossil fuel tax.
10	Dissanayake et al. [54]	Carbon Tax and ETS	Conceptual/Simulation	Energy	Carbon tax and emission trading reduce emissions, promote renewables, and generate revenue but pose minor economic challenges.
11	Dewanatan et al. [55]	Carbon Tax	Conceptual/Simulation	Industry	Carbon taxes improve project feasibility, drive emissions reductions, and require supportive incentives to mitigate economic impacts.
12	Steckel et al. [56]	Carbon Pricing	Conceptual/Simulation	Energy and Transport	Carbon pricing achieves progressive impacts with equitable revenue recycling, though outcomes vary by country and sector-specific energy use.
13	Nong et al. [57]	Carbon Tax	Conceptual/Simulation	Energy and Agriculture	Carbon tax policies effectively reduce emissions but impose higher economic costs, especially in developing countries reliant on emission-intensive inputs.
14	Mahadevan & Nugroho [58]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax paired with renewable energy investments reduces emissions, minimizes GDP loss, and mitigates poverty more effectively than gasoline taxes.
15	Hasan et al. [59]	Carbon Tax and ETS	Conceptual/Simulation	Energy	Cap and trade provides market flexibility, whereas a carbon tax directly ties emission reductions to cost—effectiveness in inventory and green technology investments.
16	Tiwari et al. [60]	Carbon Tax	Conceptual/Simulation	Transport	A carbon tax reduces logistics costs and emissions by optimizing containerization strategies and promoting greener supply chains.
17	Pertiwi et al. [61]	Carbon Pricing	Conceptual/Simulation	Forestry	Including ecosystem services in carbon pricing makes peatland conservation financially viable, supporting emissions reduction and ecosystem protection.
18	Ramadhani & Koo [35]	Carbon Tax	Conceptual/Simulation	Energy	A low-carbon tax in Indonesia mitigates carbon leakage and trade impacts from border tax adjustments while minimizing economic disruption.
19	Edianto et al. [62]	Carbon Pricing	Conceptual/Simulation	Energy	Weak carbon pricing and renewable policies drive coal financing over renewables, whereas strong carbon pricing shifts investment to cleaner energy.
20	Ordonez et al. [63]	Carbon Pricing	Conceptual/Simulation	Energy	Carbon pricing drives coal displacement, enhances renewable competitiveness, and offers fiscal benefits through revenue reinvestment.
21	Jauhari [64]	Carbon Tax	Conceptual/Simulation	Energy and Industry	Carbon taxes drive green technology adoption, reduce emissions, and improve supply chain efficiency through production adjustments and investments.
22	Tan et al. [65]	Carbon Pricing	Conceptual/Simulation	Forestry	Carbon pricing supports cost—effective peatland restoration, reduces emissions, mitigates fires, and promotes biodiversity conservation.
23	Chang & Phoumin [66]	Carbon Pricing	Conceptual/Simulation	Energy	Carbon pricing incentivizes cost—effective hydrogen production from curtailed renewables, supporting emissions reduction and renewable integration.
24	Hartono et al. [40]	Carbon Tax	Conceptual/Simulation	Energy	Carbon tax revenue recycling boosts GDP, employment, and energy efficiency, requiring targeted sectors, optimal tax levels, and diverse investments.
25	Kamandika & Dhakal [67]	Carbon Pricing	Conceptual/Simulation	Energy	Carbon pricing drives emissions reductions in Indonesia's power sector by incentivizing a transition to lower-carbon technologies.
26	Yunikewaty & Siswahyudi [68]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax reduces GHG emissions, supports green energy, and drives sustainability when combined with renewable investments and proper regulation.

Table 3 (continued)

No.	Author(s)	Policy Instrument	Implementation Phase	Sectoral Coverage	Key Focus
27	Chien et al. [69]	Carbon Tax	Conceptual/Simulation	Energy	Carbon taxes drive the low-carbon transition by reducing fossil fuel reliance, promoting renewables, and fostering green investments.
28	Hartono et al. [39]	Carbon Tax	Conceptual/Simulation	Forestry and Energy	A carbon tax in Indonesia promotes energy transformation and biofuel investment but struggles to achieve energy mix targets or deliver a double dividend.
29	Sarjiya et al. [70]	Carbon Tax	Conceptual/Simulation	Energy	Carbon taxes play a vital role in supporting renewable energy targets and coal phase-out policies in Indonesia's Java-Bali power system.
30	Setiawan et al. [44]	Carbon Pricing	Conceptual/Simulation	Energy	A carbon tax supports biomass co-firing in coal power plants by improving feasibility and incentivizing renewable energy adoption.
31	Junianto et al. [71]	Carbon Tax	Conceptual/Simulation	Industry	A carbon tax supports emissions reduction in the cement industry through clinker substitution, alternative fuels, and CCS technology adoption.
32	Langer et al. [72]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax incentivizes renewable energy adoption by enhancing the competitiveness of wind power in Indonesia's energy transition.
33	Jauhari et al. [73]	Carbon Tax	Conceptual/Simulation	Industry	A carbon tax drives green technology adoption, optimizing supply chains to balance costs, emissions, and environmental compliance.
34	Gabbar & Esteves [74]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax promotes cleaner energy in seaports, driving the adoption of nuclear-renewable systems and investments in low-carbon infrastructure.
35	Jauhari et al. [75]	Carbon Tax	Conceptual/Simulation	Industry	Carbon taxes drive green technology adoption, reduce emissions, and promote supply chain efficiency while balancing costs with incentives.
36	Suef et al. [76]	Carbon Tax	Conceptual/Simulation	Industry	A carbon tax drives hybrid production systems, optimizing supply chains to balance costs and reduce emissions.
37	Rokhmawati et al. [77]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax reduces emissions, supports renewable energy adoption, and aligns with the Environmental Kuznets Curve in Indonesia.
38	Andrianus et al. [78]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax reduces CO ₂ emissions in Indonesia but negatively impacts welfare, requiring revenue recycling and sector-specific interventions.
39	Jaya & Padilla [79]	Carbon Pricing	Conceptual/Simulation	Multi-sectoral	A carbon tax is crucial for Indonesia's green economy transition, targeting emissions-intensive sectors to align with sustainability goals.
40	Aji & Putro [80]	Carbon Tax	Conceptual/Simulation	Energy	Carbon taxes and renewable incentives reduce emissions in Indonesia's geothermal sector while accelerating sustainable energy development.
41	Rifansyah & Hakam [81]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax enhances the feasibility of floating solar photovoltaics, supporting emissions reductions and Indonesia's renewable energy transition.
42	Hersaputri et al. [82]	Carbon Tax	Conceptual/Simulation	Energy	Higher carbon tax rates in Indonesia promote emissions reductions, accelerate renewable energy adoption, and support coal phase-out efforts, contributing to a fair energy transition.
43	Rosyidi & Pratama [83]	ETS	Conceptual/Simulation	Energy	Cap-and-trade policies drive supply chain decisions toward minimizing emissions while balancing cost efficiency and environmental sustainability.
44	Apriliyanti et al. [84]	Carbon Pricing	Conceptual/Simulation	Energy	A carbon tax, alongside market reforms, promotes renewable energy adoption but is hindered by opposition from established coal industry stakeholders.
45	Rahmanta et al. [45]	Carbon Tax	Conceptual/Simulation	Energy	A carbon tax encourages biomass co-firing and CCS adoption, lowering emissions while improving the economic viability of low-carbon technologies.
46	Yola et al. [85]	Carbon Pricing	Conceptual/Simulation	Transport	A carbon tax, combined with ecospatial monitoring, helps reduce traffic emissions, promotes transportation reforms, and enhances sustainable urban planning.

based group, which predominantly explores theoretical frameworks and model-based assessments using tools such as CGE and energy system models. The carbon tax and the ETS are the most frequently analyzed instruments in this group, often examined alongside complementary measures such as energy subsidy reform or CCS. The energy sector dominates these analyses, reflecting its centrality in Indonesia's emissions profile and mitigation strategies.

The pilot or early-stage implementation group focuses on case studies and regional demonstration initiatives, especially in forestry. REDD+ and voluntary carbon markets are the dominant instruments in this group, highlighting the emphasis on nature-based solutions. Although energy and transport sectors are also represented, the forestry sector remains prominent due to its substantial mitigation potential and international support mechanisms.

The legislated group, though smaller, illustrates a transition toward formal regulation. This group features studies addressing the incorporation of carbon pricing into national policy, including hybrid mechanisms such as carbon pricing with the CBAM. Energy and industry sectors are

the primary focus, aligned with Indonesia's policy priorities under the CEV framework.

Overall, this body of literature reflects a ladder-type evolution from conceptual understanding to pilot testing and formal policy adoption. This pattern highlights the importance of integrating academic research, pilot initiatives, and regulatory developments to enhance the design and implementation of carbon pricing mechanisms in Indonesia.

To deepen this observation, the findings were examined in relation to existing literature and recent policy developments. The synthesis reveals a clear dominance of theoretical or model-based research (over 65%), particularly within the energy sector. While these studies offer valuable insights into the cost-effectiveness, equity implications, and technical feasibility of carbon pricing instruments in Indonesia, policy implementation lags considerably behind analytical recommendations. For instance, simulation-based studies frequently propose carbon tax rates between 30–60 USD/tCO₂ to stimulate renewable energy deployment and deliver economic co-benefits [38–40], yet the current rate remains at 2 USD/tCO₂ with limited coverage and delayed enforcement. This

Table 4

Classification of carbon pricing articles in the legislated group.

No.	Author(s)	Policy Instrument	Implementation Phase	Sectoral Coverage	Key Focus
1	Roelfsema et al. [86]	Carbon Tax and ETS	Legislated	Energy	ETEs reduce emissions, foster innovation, and align national efforts with global climate targets.
2	Stevens [87]	Carbon Tax and ETS	Legislated	Energy and Industry	Emission-intensive industries reduce the stringency of both ETEs and carbon taxes, with policy impacts varying by sector and activity.
3	Dyarto & Setyawan [41]	Carbon Tax	Legislated	Multi-sectoral	Political resistance and lack of coordination hinder carbon tax implementation in Indonesia, despite its potential to reduce emissions and fund climate programs.
4	Putra et al. [88]	Carbon Tax and ETS	Legislated	Energy	A carbon tax offers simplicity in implementation, whereas cap and trade provides cost-effective emissions control through market trading.
5	Sasaki [89]	Carbon Tax	Legislated	Forestry	Carbon taxes and reduced-impact logging reduce emissions, promote bioenergy, and generate significant revenue for sustainable forest management.
6	Sasaki et al. [90]	Carbon Tax	Legislated	Forestry	A carbon tax supports emissions reductions and sustainable forest management in Southeast Asia through REDD+ activities.
7	Cetera [91]	Carbon Trading	Legislated	Forestry	Carbon trading in Indonesia supports sustainable forest management but requires clearer carbon rights and conflict resolution.
8	Kamil et al. [42]	Carbon Tax	Legislated	Energy	A carbon tax in Indonesia generates significant revenue, boosts regional economies through revenue recycling, and encourages low-carbon technologies.
9	Leonard et al. [92]	Carbon Tax	Legislated	Energy	A carbon tax generates revenue to support renewables and green investments, requiring improved regulations for sustainable development.
10	Daryanto & Setyanto [93]	ETS	Legislated	Energy and Industry	Carbon caps and prices drive emissions management strategies by optimizing production and minimizing costs.
11	Syahrudin et al. [94]	Carbon Trading	Legislated	Multi-sectoral	Carbon trading in Indonesia supports emissions reduction and sustainable investment through market-driven mechanisms and robust governance.
12	Wiloso et al. [7]	Carbon Pricing	Legislated	Energy and Industry	Carbon taxes and ETEs are vital for mitigating emissions, preventing carbon leakage, and promoting Indonesia's green supply chain transition.
13	Nurhayati et al. [95]	Carbon Tax and ETS	Legislated	Energy	Indonesia's cap and tax scheme uses carbon pricing to reduce emissions, promote green technologies, and support sustainable economic growth.
14	Wirawan et al. [96]	Carbon Tax	Legislated	Energy and Transport	A carbon tax supports Indonesia's biodiesel program by internalizing fossil fuel costs, promoting cleaner energy, and advancing sustainability.

Table 5

Classification of carbon pricing articles in the pilot/early-stage group.

No.	Author(s)	Policy Instrument	Implementation Phase	Sectoral Coverage	Key Focus
1	Jaenicke et al. [97]	Carbon Trading	Pilot/Early-Stage	Forestry	Carbon trading supports peatland restoration by monetizing emission reductions, enabling climate mitigation and biodiversity conservation.
2	Lu & Liu [98]	Carbon Trading	Pilot/Early-Stage	Forestry	REDD+ carbon trading facilitates emissions reductions by incentivizing deforestation avoidance but depends on higher prices and fair policies for effectiveness.
3	Hilwan et al. [99]	Carbon Trading	Pilot/Early-Stage	Forestry	Carbon trading through REDD+ supports sustainable forest management by generating carbon credits, reducing emissions, and offering financial incentives for climate mitigation efforts.
4	Nguyen et al. [100]	ETS	Pilot/Early-Stage	Energy	A regional ASEAN ETS reduces costs for permit buyers but challenges permit sellers, requiring equitable revenue recycling for balanced outcomes.
5	Hasan et al. [101]	Carbon Tax and ETS	Pilot/Early-Stage	Energy	A carbon tax directly reduces emissions, whereas cap and trade offers cost-effective compliance, both benefiting from green technology investments.

divergence between modeled scenarios and real-world application reflects institutional fragmentation and political resistance, as highlighted in legislated studies [41,42].

Moreover, although many conceptual studies emphasize integrating carbon pricing with renewable energy and CCS technologies [43–45], practical adoption is constrained by financial, regulatory, and governance barriers. Bibliometric clustering reinforces this mismatch, revealing strong academic interest in energy transition themes but limited focus on MRV systems, revenue recycling, and cross-sectoral coordination in practice. Comparative analysis in Section 4.2 shows that mature systems achieve closer alignment between research, policy, and measurable outcomes. Bridging this gap in Indonesia will require strengthening institutional capacity, improving interagency coordination, and fostering stakeholder engagement, thereby providing a critical foundation

for deeper examination of their implications for policy design, governance arrangements, and implementation strategies, while also offering an evidence-based basis for aligning future policy reforms with empirical insights from scholarly research.

4. Discussion

4.1. Current status of carbon pricing in Indonesia

As the world confronts the escalating impacts of climate change, Indonesia, one of the largest global CO₂ emitters, has taken critical steps toward mitigating its environmental footprint [102]. In 2022, global CO₂ emissions reached 57.4 gigatons, with Indonesia making a significant contribution to this total. The country saw its emissions rise from

615.92 million tons in 2021 to 728.88 million tons in 2022, primarily driven by electricity and heat production, agriculture, and transportation [103]. Recognizing its pivotal role in global emissions, Indonesia has committed to achieving net zero emissions by 2060, aiming to balance its development needs with sustainability goals. Central to this commitment are carbon pricing mechanisms, including carbon taxes and emissions trading, which internalize the costs of emissions and encourage low-carbon practices across sectors. Indonesia's carbon pricing strategy is embedded within a broader framework of climate policy commitments and is aligned with global initiatives such as the Paris Agreement. The LTS-LCCR 2050 and ENDC prioritize reducing emissions in key sectors, particularly energy, which remains Indonesia's largest contributor to CO₂ emissions. Several legislative milestones underpin these efforts, including Law 16/2016, which ratifies the Paris Agreement, and Law 7/2021, which incorporates a carbon tax into Indonesia's tax regulations. Presidential Regulation 98/2021 further defines the economic value of carbon, providing a legal foundation for carbon pricing mechanisms. Together, these frameworks represent a systematic approach to addressing emissions while fostering sustainable growth.

A significant component of Indonesia's climate strategy is its ETS, which was introduced as a voluntary, intensity-based pilot program in 2021 to target the power generation sector [104]. This pilot phase aimed to assess the feasibility of ETS implementation by assigning emissions intensity targets to coal-fired power plants. On February 22, 2023, the MEMR, officially launched a mandatory ETS for the power sector, requiring plants to comply with specific intensity limits [105]. Under this system, facilities exceeding their limits must purchase carbon credits from those operating below their targets or face penalties, including a carbon tax. Initially covering 99 coal-fired power plants, the mandatory ETS is expected to expand to 146 plants by 2024, demonstrating its growing role as a key mechanism in Indonesia's low-carbon transition [106].

However, as the ETS expands, it faces several critical challenges that must be addressed to ensure its effectiveness. Transparency and credibility are essential, requiring robust verification of emissions data, as well as the availability of reliable carbon credits, to build stakeholder trust. Maintaining market stability by balancing the supply and demand of credits is crucial to avoiding price volatility, which could undermine the system's functionality [107,108]. Additionally, the ETS's growth demands enhanced regulatory capacity to monitor compliance, enforce rules, and prevent market manipulation. A centralized platform for emissions tracking and a transparent regulatory framework are necessary to ensure equitable stakeholder participation. Without these measures, the ETS risks falling short of its potential to drive substantial emissions reductions [109].

Complementing the ETS is Indonesia's carbon tax, introduced under Law 7/2021. The tax is set at an initial rate of 30 IDR per kilogram of CO₂, equivalent to 2 USD per tonne of CO₂, which is significantly lower than the international recommendations of 30-100 USD per tonne of CO₂ for developing countries [110,111]. The low initial rate reflects Indonesia's economic realities, but it may not provide sufficient incentives for industries to transition to cleaner technologies or meaningfully reduce emissions. Originally scheduled for implementation in 2022, the tax rollout has been delayed to 2025, highlighting the complexities of introducing such mechanisms in a developing economy [112]. This delay also underscores the challenges associated with disaster relief, considering the impacts of the COVID-19 pandemic, and balancing the economic recovery with ambitious climate policy goals.

Despite its modest start, the carbon tax has significant potential to evolve into a robust emissions reduction tool. Gradual increases in the tax rate can align Indonesia's pricing with global benchmarks, thereby providing stronger incentives for businesses to adopt sustainable practices [113]. Moreover, the generated revenue offers opportunities for reinvestment in renewable energy, energy efficiency programs, and climate resilience initiatives [114]. Transparent communication about how these funds will be allocated is essential for building trust among

stakeholders and ensuring public support. By strategically utilizing carbon tax revenues, Indonesia can accelerate its low-carbon transition while addressing pressing social and environmental needs [92].

The interplay between the carbon tax and the ETS presents an opportunity for a more integrated and effective climate strategy. While the trading system enables flexibility through cost-effective emissions reductions, the carbon tax provides a predictable price signal to guide long-term investments in cleaner technologies [115,116]. Together, these mechanisms can complement each other to enhance emissions reductions across various sectors. However, achieving this synergy requires careful coordination between policymakers, regulators, and industries. Clear guidelines on their interaction and evolution over time are crucial for avoiding redundancies and ensuring maximum impact [13].

Indonesia's energy sector, which is heavily reliant on coal, remains central to its carbon pricing initiatives. Coal-fired power plants are a significant source of CO₂ emissions, making them the primary focus of both the carbon tax and ETS. The government's efforts to introduce emissions caps and incentivize renewable energy alternatives reflect a commitment to addressing these challenges. However, the pace of transitioning from coal to renewables has been slow. Despite its vast renewable energy potential, including over 3294.4 GW in solar, 154.9 GW in wind, and 23 GW in geothermal [9], financial, technical, and regulatory barriers have hindered progress. Accelerating this transition will be critical for meeting Indonesia's climate targets.

To ensure the success of its carbon pricing mechanisms, Indonesia must address several overarching challenges. Strengthening data validation and developing a transparent monitoring system are vital for maintaining confidence in the carbon market. Gradually increasing the carbon tax rate while balancing economic impacts can provide stronger incentives for industries to reduce emissions [117]. Additionally, building institutional and technical capacity will be essential to managing the complexities of an expanding carbon market. Finally, fostering broad stakeholder engagement and ensuring equitable participation will enhance the credibility and acceptance of these policies, particularly among the industries and communities most affected by the transition [118].

Indonesia's carbon pricing strategy, encompassing both the carbon tax and ETS, represents a crucial step in its journey toward achieving net zero emissions by 2060. The expansion of the ETS to 146 power plants by 2024 and the planned implementation of the carbon tax reflect the government's commitment to climate action. Nevertheless, several challenges, such as delays in implementation, low initial tax rates, and market inefficiencies, highlight the need for continuous refinement and adaptation. By addressing these obstacles and leveraging its vast renewable energy potential, Indonesia can position itself as a regional leader in carbon pricing and contribute meaningfully to global sustainability efforts.

In summary, the findings of this review in response to RQ1 suggest that Indonesia's carbon pricing approach reflects a pragmatic balance between emissions reduction and sustainable development. The gradual introduction of the carbon tax and ETS demonstrates an intent to integrate climate policy without disrupting economic stability [58,40]. While institutional and technical limitations remain, these mechanisms form foundational steps toward a low-carbon pathway [119,120].

Several studies support the technical feasibility of this transition, particularly through carbon-pricing-induced investments in biomass co-firing in coal-fired power plants, geothermal expansion, and floating solar PV deployment [44,80,81]. These renewable technologies are abundant in Indonesia and offer a strategic opportunity to align fiscal instruments with long-term decarbonization. Recent global reviews have underlined the transformative role of renewable energy technologies, including solar, wind, tidal, and geothermal, in advancing sustainable energy transitions and supporting social equity goals [121]. In global contexts, the deployment of low-carbon technologies, including biomass, hydrogen, and CCS, is increasingly framed within sustainable

energy systems supported by robust pricing signals and regulatory incentives [122].

Additionally, carbon tax revenue recycling has been shown to enhance energy access, stimulate GDP growth, and improve social equity outcomes, reinforcing the synergy between climate action and development objectives. The effectiveness and legitimacy of such instruments are further strengthened when embedded within coherent legal and regulatory frameworks that support circular economy transitions and low-carbon innovation [123].

With enhanced policy coordination, scaled-up technology deployment, and transparent governance, carbon pricing mechanisms can evolve into a cornerstone of Indonesia's climate-neutral development trajectory. A broader understanding of how Indonesia's framework compares to global practices provides further insight into potential pathways for strengthening and scaling its carbon pricing strategy, as explored further in the subsequent subsection on comparative insights.

4.2. Indonesia's carbon pricing framework in a global context

Carbon pricing has increasingly emerged as a pivotal instrument in the global endeavor to mitigate GHG emissions, offering a market-based framework that seeks to internalize the external costs of carbon pollution while simultaneously driving the transition toward a low-carbon economy [124]. Within this global context, developed economies have spearheaded the implementation of advanced systems characterized by high carbon tax rates and comprehensive ETS frameworks, both of which have delivered measurable and sustained reductions in emissions. In contrast, emerging economies—Indonesia among them—remain at relatively early stages of adoption, reflecting the interplay between unique macroeconomic structures, institutional capacities, and socio-economic priorities. The current architecture of Indonesia's carbon pricing framework, composed of a modestly priced carbon tax alongside an emergent ETS, reflects a deliberately cautious but strategically significant approach to balancing national development imperatives with climate change mitigation commitments.

This subsection directly addresses RQ2, which examines the comparative position of Indonesia's carbon pricing architecture relative to international experiences, with a view to extracting practical lessons that may inform both domestic policy refinements and broader global carbon pricing strategies. By situating Indonesia's framework within the wider landscape of global practices, the analysis seeks to identify both the systemic challenges and incremental progress that characterize the country's current trajectory. Furthermore, the discussion aims to distill actionable insights that can guide Indonesia in refining its mechanisms while offering transferrable lessons to other emerging economies. In doing so, the comparative assessment underscores Indonesia's potential to contribute to regional climate governance while aligning national mechanisms with evolving global norms.

Table 6 synthesizes comparative information on carbon pricing systems implemented across a spectrum of national contexts, revealing notable variation in scope, ambition, and implementation effectiveness. Countries such as Sweden, Switzerland, and Canada have embedded high-rate taxes and mature ETS frameworks that collectively enable significant emissions abatement. Conversely, many developing countries maintain lower price signals or narrower coverage, constrained by fiscal limitations and institutional readiness. Indonesia aligns with this latter category, as it attempts to reconcile a heavy reliance on coal and sustained economic expansion with the long-term objective of reducing GHG emissions. These observed disparities in ambition and capacity illuminate valuable considerations for tailoring carbon pricing systems to specific national circumstances.

Experiences from developed economies highlight the transformative potential of robust, well-calibrated carbon pricing instruments. Sweden, for instance, has progressively increased its carbon tax to 127.26 USD per tonne of CO₂ by 2024, while Switzerland's carbon tax, at 132.12 USD per tonne, ranks among the highest globally [125]. These pricing mechanisms

are underpinned by stringent institutional frameworks, targeted incentives for renewable energy deployment, and structured strategies for public engagement to ensure political acceptability. The EU ETS, launched in 2005, offers an exemplary model of a cap-and-trade mechanism that operates under ambitious annual reduction trajectories, encompassing power generation, industrial processes, and aviation [126]. Collectively, these examples illustrate how ambitious pricing, combined with robust governance and broad sectoral inclusion, can produce meaningful and sustained mitigation outcomes.

Canada's approach provides an instructive case of policy hybridity, blending a federal-level carbon tax with subnational ETS initiatives. The federal tax, introduced at 20 CAD per tonne in 2019, is legislated to rise incrementally to 170 CAD by 2030 [142]. To preserve public trust and mitigate distributional impacts, Canada redistributes a portion of tax revenues through household rebates, providing near-term economic relief while maintaining long-term climate incentives. At the regional level, Quebec operates an ETS linked to California's system, thereby demonstrating the feasibility and benefits of cross-jurisdictional cooperation. In contrast, many emerging economies contend with constrained fiscal space, weaker administrative capacity, and entrenched fossil fuel dependencies. For example, Chile and Colombia have both adopted modest carbon taxes of 5 USD per tonne [125], laying important policy foundations without yet delivering transformative change. Colombia's intent to integrate an ETS with its tax contrasts with Chile's absence of a trading mechanism, underscoring the institutional prerequisites of market-based schemes.

Additional country cases offer further perspective on the design and implementation of carbon pricing. Brazil has yet to legislate a national carbon tax but is progressing toward a voluntary ETS aligned with REDD+ initiatives and national land-use strategies, reflecting its comparative advantage in nature-based solutions. India, another coal-dependent economy, has opted for a coal cess and the PAT scheme—while not a conventional ETS, PAT demonstrates how sector-specific instruments can evolve under political and market constraints. Vietnam, still in the preparatory phase of its ETS, benefits from targeted international support through the PMR and the UNDP, highlighting the catalytic role of external assistance in building institutional and technical readiness.

South Africa and Singapore represent distinct models of gradualism in carbon pricing. South Africa introduced its carbon tax in 2019 at a relatively low rate of 7 USD per tonne, extending coverage to energy, industrial, and transport sectors [151]. While the initial price signal is modest, the breadth of sectoral inclusion marks an important step toward more comprehensive frameworks. Singapore, classified as a high-income economy, implemented a carbon tax of 3.69 USD per tonne in 2019, targeting high-emitting installations. This rate is scheduled to increase to 18.48 USD per tonne by 2024, with coverage spanning the electricity, industrial, and waste sectors. The predictable escalation and multi-sector coverage illustrate how measured price pathways can maintain economic competitiveness while progressively advancing climate objectives [152].

Within the ASEAN context, Table 6 underscores marked differences in policy maturity. Singapore maintains a clearly articulated tax trajectory, positioning itself as the regional leader in carbon pricing. Indonesia, by contrast, has initiated a carbon tax set at 2 USD per tonne and launched an ETS in 2023 focusing primarily on coal-fired power plants. Vietnam remains at the conceptual stage, with no operational tax or trading scheme. These variations reflect broader differences in institutional capacity, sectoral priorities, and political economy considerations among ASEAN member states.

Beyond national frameworks, international cooperation plays an increasingly strategic role in shaping the ambition and architecture of carbon pricing. Indonesia's engagement through the JETP and preparatory measures for compliance with the CBAM signal a gradual alignment with evolving climate finance mechanisms and international trade regulations. South Africa, similarly, has integrated its carbon tax within a

Table 6

Overview of carbon tax and ETS implementation in selected countries, with emphasis on policy-economy fit, governance impacts, and international cooperation strategies.

Country	Carbon Tax System	ETS System	Policy-Economy Fit	Governance Impacts	International Cooperation Strategies	Ref.
Indonesia	Proposed at 30,000 IDR (2 USD)/tCO ₂ e; implementation delayed.	Implemented 2023; free allocation in power sector	Coal-dominant; low carbon price	MoF-MoEF-MEMR coordination on fiscal, MRV, and power sector implementation	Linked to JETP and CBAM response	[11,125,127]
South Africa	Established in 2019; 10.09 USD/tCO ₂ e by 2024.	No national ETS	Coal-reliant; phased design	National Treasury-LED	Green recovery and Just Energy Transition	[128,129]
Brazil	No national carbon tax	Voluntary ETS under development	REDD+ aligned with land policy	REDD+ institutions strong	Active in UNFCCC; developing MRV	[130,131]
India	No explicit tax; coal cess used	No national ETS; PAT scheme in place	Coal-dependent; political resistance	Central coordination, state implementation	Carbon intensity target; CBAM preparation	[132–134]
Vietnam	No national carbon tax	ETS under development	Rapid industrialization	Cross-ministerial coordination	Supported by PMR and UNDP	[135,136]
Colombia	Started 2017; 6.68 USD/tCO ₂ e in 2024	ETS planned alongside tax and offsets	Blended tax-offset fits commodity economy	National carbon office	Leader in nature-based solutions in Latin America	[137,138]
Finland	First in 1990; now 99.99 USD/tCO ₂ e	EU ETS Phase IV	Tax aligns with low-carbon economy	Integrated EU-level planning	EU climate leadership	[139,140]
Sweden	Since 1991; 127.26 USD/tCO ₂ e	EU ETS	Integrated with energy and welfare policies	Strong institutional design	Aligned with EU climate targets	[139,140]
Norway	Since 1991; 107.78 USD/tCO ₂ e	EEA-linked to EU ETS	Oil exporter; reinvests revenue	Dual system (tax + ETS)	High transparency; cross-border trading	[139,140]
Denmark	Since 1992; 28.21 USD/tCO ₂ e	EU ETS	Supports renewable shift	Stable national governance	Fully integrated with EU ETS [139,140]	
Switzerland	Since 2008; 132.12 USD/tCO ₂ e	Linked to EU ETS	Tax with household rebate	Independent but aligned	Cross-border linkage for ETS	[139,141]
Canada	Federal backstop; 58.95 USD/tCO ₂ e in 2024	Provincial ETS (e.g., Quebec)	Federal-provincial balance	Pan-Canadian framework	CBAM-responsive; NAFTA partners	[142–144]
Chile	Since 2017; 5 USD/tCO ₂ e	ETS in pilot stage	Low rate fits gradual shift	Green tax reform institution	Pacific Alliance carbon market	[130,145]
Japan	Since 2012; 2.65 USD/tCO ₂ e	Tokyo/Saitama ETS	Low tax complements efficiency policy	Local-level ETS leadership	Voluntary carbon market with MRV	[146–148]
Singapore	Since 2019; rising to 25 SGD/tCO ₂ e	No national ETS	Aligned with green economy goals	Carbon pricing act; strong MRV	Positioned as carbon services hub	[149,150]

broader just transition framework supported by green recovery funds and multilateral financing [129]. Brazil's ETS development is embedded within its REDD+ strategy, while Colombia focuses on enhancing the integrity of carbon offsets in Latin America [130]. Vietnam's ETS roadmap benefits from capacity-building support for MRV systems through PMR and UNDP [135]. Canada's carbon pricing aligns with CBAM benchmarks and maintains policy coherence with its NAFTA partners, whereas Chile participates in the Pacific Alliance carbon market initiative [145].

Linking mechanisms between national carbon pricing systems can enhance liquidity, price stability, and overall market efficiency. The EU–Switzerland ETS linkage is a prime example, demonstrating how mutual recognition of allowances, harmonized monitoring protocols, and coordinated policy adjustments can yield efficiency gains for both markets. Similarly, the linkage between Quebec and California under the Western Climate Initiative [153] illustrates that such integration can function effectively even across international borders with different governance systems. These linkages not only strengthen market depth but also promote cross-border investment in low-carbon technologies. However, linking requires significant alignment in allocation methods, compliance rules, and enforcement capacity [154]—challenges that can be formidable for developing economies with nascent systems.

Global lessons from ETS design also highlight the importance of ambitious cap trajectories, stringent monitoring and verification procedures, and robust compliance enforcement. The EU ETS operates with a linear reduction factor that tightens caps annually, creating a predictable scarcity of allowances and maintaining upward pressure on

carbon prices [126]. China's national ETS, though the largest globally in terms of emissions coverage, initially faced significant challenges including over-allocation, lack of sectoral diversity, and limited trading activity [155]. Addressing these constraints has involved iterative reforms, such as tightening benchmarks and introducing more sectors into the system. These adjustments illustrate that policy learning and gradual strengthening of market rules are central to maintaining ETS credibility.

For Indonesia, these experiences underline the importance of designing an ETS that avoids excessive initial allocation and incorporates clear plans for cap tightening. Over-allocation, as seen in China's early stages, can suppress price signals and diminish incentives for emissions reduction. Similarly, ensuring sectoral diversity—by progressively including industries beyond the power sector—can prevent market concentration and enhance resilience to sector-specific shocks. International examples demonstrate that a phased inclusion strategy, combined with robust MRV systems, can facilitate a smooth transition from pilot schemes to a mature market. The strategic placement of Indonesia's ETS within its broader decarbonization framework will determine its long-term relevance and effectiveness.

Another critical dimension of ETS effectiveness lies in the behavior of allowance prices, which can be highly sensitive to macroeconomic conditions, policy uncertainty, and external shocks. Price volatility can undermine investment planning, especially in capital-intensive low-carbon technologies [156]. In the EU ETS, market stability reserve mechanisms have been deployed to address such volatility by adjusting auction volumes in response to market imbalances [157]. China's ETS, lacking a comparable mechanism in its initial years, experienced prolonged pe-

riods of low trading volumes and stagnant prices. These contrasting experiences point to the necessity of embedding stabilizing instruments early in ETS design.

Price predictability can also be enhanced through the use of advanced forecasting tools. In the EU, machine learning models—including CNN—have been applied to anticipate short-term price movements, thereby informing both market participants and regulators [158]. While such predictive tools cannot eliminate volatility, they can improve decision-making and reduce exposure to sudden price swings. In parallel, agent-based modeling has been employed in Shanghai's ETS to simulate market participant behavior under different policy scenarios, revealing that a balanced mix of penalties and incentives is most effective in driving low-carbon investment [159]. These analytical approaches underscore the value of integrating data-driven insights into carbon market governance.

The relationship between carbon pricing and technological innovation is another domain where international experience offers valuable insights. In China's thermal power sector, for example, research indicates that significant low-carbon innovation tends to occur only once carbon prices surpass certain threshold levels [160,161]. This suggests that price signals must be sufficiently strong to justify the capital costs and operational risks associated with new technologies. In the EU ETS, complementary policies such as renewable energy subsidies and research grants have been essential [162] in reinforcing the innovation stimulus provided by carbon pricing. Without such complementary measures, price signals alone may not be adequate to catalyze substantial technological change.

Indonesia can draw upon these lessons by ensuring that its carbon pricing system—whether tax-based, market-based, or hybrid—establishes clear price trajectories that reach levels capable of influencing investment decisions. A modest initial rate may ease political acceptance, but the absence of a defined path toward higher prices could weaken the system's transformative potential. Furthermore, integrating carbon pricing with targeted support for renewable energy, energy efficiency, and industrial decarbonization could create a reinforcing policy environment. International evidence suggests that this policy coherence is a critical determinant of long-term emissions reduction outcomes.

Sectoral readiness is equally critical in determining how carbon pricing impacts firm behavior and broader market dynamics. In China, studies have found that ETS participation has heterogeneous effects on firm value depending on a company's financial structure, sectoral competitiveness, and prior exposure to environmental regulations [163]. In some sectors, carbon pricing has prompted operational efficiency gains and stimulated low-carbon product development. In others, especially where compliance costs are high and technology options are limited, firm value has been negatively affected. These findings reinforce the importance of tailoring compliance flexibility, such as allowing for offset use, while ensuring environmental integrity.

Macroeconomic linkages also shape ETS outcomes. Research on price co-movements between the EU and China indicates that carbon markets can exhibit correlated volatility driven by global energy prices, exchange rate fluctuations, and synchronized economic cycles [164,165]. This interconnectedness implies that shocks in one major market can ripple through others, affecting allowance prices and potentially undermining domestic policy objectives. For emerging markets, understanding these linkages is crucial for designing safeguards that protect domestic market stability without isolating the system from beneficial international trends.

Beyond technical and market considerations, public acceptance and political stability remain decisive factors in sustaining carbon pricing mechanisms. Experiences from France's "gilets jaunes" protests, which were partly triggered by fuel tax increases, underscore the risks of neglecting distributional impacts [166,167]. Successful systems tend to recycle revenues toward visible public benefits [168], such as household rebates, public transportation investments, or clean energy subsidies. Transparent communication about the use of carbon pricing revenues

can also strengthen public trust, which in turn supports policy durability.

International cooperation, particularly in the form of knowledge-sharing platforms and technical assistance programs, has proven valuable in accelerating learning curves for new carbon pricing adopters. Initiatives such as the World Bank's Partnership for Market Implementation and the International Carbon Action Partnership [169,154] facilitate peer-to-peer exchange, capacity building, and the dissemination of best practices. These networks can help emerging economies anticipate common challenges, adapt proven solutions, and avoid costly policy missteps.

For Indonesia, active participation in such international networks can complement domestic policy development by providing access to both technical expertise and practical policy experience. This external engagement is especially valuable during the transition from pilot programs to fully operational national mechanisms. In addition to benefiting from global lessons, Indonesia's own experience—particularly in integrating carbon pricing with broader energy transition policies—can contribute to the collective knowledge base, positioning the country as a constructive actor in regional and global climate governance.

In summary, the comparative review reveals that while Indonesia's carbon pricing framework is at an early stage, the international evidence base offers a wealth of transferable lessons. These include the need for clear and escalating price trajectories, robust governance structures, mechanisms for price stability, and the integration of carbon pricing with complementary sectoral policies. As Indonesia advances from pilot initiatives to a fully developed system, careful adaptation of global best practices—tempered by local socio-economic realities—will be essential to ensuring both environmental effectiveness and economic resilience.

5. Conclusion

Indonesia's carbon pricing framework has been evaluated through a global lens, using systematic literature review, bibliometric mapping, and comparative analysis to examine the role of the carbon tax and emissions trading in its climate transition. The assessment identified key aspects of policy design, institutional coordination, sectoral coverage, and international alignment, situating Indonesia's approach within broader global trends. Current initiatives, comprising a modest carbon tax and a pilot ETS, mark early yet significant steps toward embedding market-based climate instruments into national development planning. However, persistent challenges—including limited sectoral coverage, low price signals, and institutional fragmentation—underscore the need for a more ambitious, flexible, and coordinated framework.

The Indonesian case offers broader lessons for other developing economies. The combination of a carbon tax and ETS provides a pragmatic pathway to balance climate objectives with development priorities, particularly in contexts of high fossil fuel dependence and constrained institutional capacity. Integrating carbon pricing with low-carbon technologies such as biomass co-firing, geothermal power, and floating solar demonstrates scalable solutions grounded in domestic resources. From a theoretical perspective, Indonesia's experience enriches understanding of how political economy dynamics, sequencing strategies, and hybrid instruments shape carbon pricing in developing contexts, contributing to the literature on adaptive climate policy design in the Global South.

To enhance policy effectiveness, three actionable pathways emerge from global practice. First, a graduated carbon tax roadmap could target an increase from 2 USD to 10–15 USD per tonne by 2030, aligned with international benchmarks and domestic socioeconomic indicators. Second, phased ETS expansion can proceed in three stages: 2023–2025 for coal-fired power plants, 2026–2028 for industry and transport, and 2029–2032 for forestry and agriculture offsets, enabling institutional readiness and market adjustment. Third, operational frameworks for ASEAN collaboration should harmonize MRV standards, develop interoperable emissions registries, and design equitable benefit-sharing

schemes to support lower-capacity members. Drawing from Canada's federal-provincial model and EU ETS integration, Indonesia can position itself as a regional leader in shaping a collaborative carbon pricing bloc in Southeast Asia.

For policymakers, actionable priorities include: (i) publishing a clear multi-year price corridor for both the carbon tax and ETS; (ii) strengthening MRV through standardized digital reporting formats, risk-based third-party verification, and public facility-level disclosure; (iii) recycling revenues transparently to protect vulnerable groups and finance grid upgrades, energy storage, efficiency measures, industrial electrification, and MRV capacity; and (iv) adopting basic market-stabilization safeguards. For industry stakeholders, priorities include: (i) adopting internal shadow prices aligned with the announced corridor; (ii) prioritizing high-return abatement options such as efficiency improvements, waste heat recovery, fuel switching, and process control; and (iii) preparing to scale medium-term measures including biomass co-firing, geothermal expansion, and floating solar PV, supported by digital MRV and voluntary disclosure.

Looking ahead, future-proofing carbon pricing will depend on deeper digitalization, transparency, and integration of advanced analytics. Standardized plant-level telemetry, satellite-assisted monitoring for land-based activities, and near real-time public dashboards can enhance compliance and benchmarking. Emerging technologies such as artificial intelligence and machine learning can strengthen carbon pricing systems through predictive analytics for allowance prices, anomaly detection in MRV datasets, optimization of allowance allocation, and adaptive policy calibration. Interoperable registries and harmonized methodologies across ASEAN will reduce transaction costs and enable market coupling, while practical support for trade-exposed sectors can improve readiness for evolving border measures. Together, predictable pricing, broader coverage, high-integrity MRV, disciplined revenue use, technological innovation, and regional cooperation can transform Indonesia's initial steps into a robust and enduring carbon pricing architecture, offering a replicable model for emerging economies aligning economic growth with net-zero commitments.

CRediT authorship contribution statement

Wahri Sunanda: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **M. Isaenaeni Bambang Setyonegoro:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Sasongko Pramono Hadi:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation. **Sarjiya:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Ethical statement

The paper has been submitted with full responsibility, following due ethical procedure, and there is no duplicate publication, fraud, plagiarism. None of the authors of this paper has a financial or personal relationship with other people or organizations that could inappropriately influence or bias the content of the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data sources are publicly available and have been appropriately cited in the paper.

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